Operational Equity Absorption and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Operational Equity Absorption (OEA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on OEA achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 32 (30) bps/month with a t-statistic of 3.49 (3.37), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing) is 28 bps/month with a t-statistic of 3.22.

1 Introduction

Financial markets exhibit systematic patterns in the cross-section of expected returns that challenge traditional asset pricing models. While extensive research documents numerous return predictors, the economic mechanisms driving these patterns remain contentious, particularly regarding whether they reflect risk compensation or market inefficiencies. The production-based asset pricing literature offers a compelling framework for understanding cross-sectional return variation through firms' real investment decisions and operating constraints. Firms face heterogeneous adjustment costs and investment frictions that create time-varying exposures to systematic productivity shocks, generating predictable variation in required returns. Despite theoretical advances linking production decisions to asset prices, empirical evidence connecting specific operating characteristics to risk premia through production-based channels remains limited.

We propose that Operational Equity Absorption (OEA) captures firms' marginal cost of capital and investment flexibility in a production economy, thereby predicting cross-sectional variation in expected returns. In production-based models, firms with higher operational equity absorption face steeper adjustment costs when responding to productivity shocks, as documented by Zhang (2005) and Liu et al. (2009). These firms exhibit lower investment-capital sensitivity and reduced operational flexibility, characteristics that Cooper and Priestley (2011) associate with higher systematic risk exposure. The mechanism operates through the q-theory channel where firms' investment decisions reflect the shadow value of capital, and heterogeneous adjustment costs create cross-sectional dispersion in conditional risk premia (Cochrane, 1991; Berk et al., 1999).

Operational equity absorption directly measures the extent to which firms' equity capital becomes locked into ongoing operations, creating investment irreversibility that amplifies exposure to aggregate productivity shocks. Following Zhang (2005), firms with high OEA face convex adjustment costs that prevent optimal capital reallocation during economic downturns, increasing their systematic risk. This friction manifests as higher required returns to compensate investors for bearing undiversifiable productivity risk. The production-based framework of Li et al. (2009) predicts that firms constrained in their ability to adjust operating leverage exhibit stronger covariation with aggregate investment shocks, a primary source of systematic risk in production economies.

The cross-sectional pricing implications emerge through the interaction between operational constraints and time-varying risk premia. High OEA firms experience greater sensitivity to discount rate shocks because their inflexible cost structures amplify cash flow volatility during productivity fluctuations (Gomes and Schmid, 2010). These firms cannot easily scale operations in response to demand shocks, creating operating leverage that Carlson et al. (2004) show increases systematic risk exposure. Moreover, the investment-based asset pricing model of Lin and Zhang (2013) demonstrates that firms with higher marginal costs of investment command higher expected returns, precisely the characteristic captured by operational equity absorption. The production-based risk channel thus predicts a positive relation between OEA and expected returns, reflecting rational compensation for exposure to systematic productivity and investment shocks.

Our empirical analysis reveals that Operational Equity Absorption strongly predicts cross-sectional stock returns. The value-weighted long/short OEA strategy achieves an annualized gross Sharpe ratio of 0.48, with monthly average returns of 34 basis points (t-statistic = 3.74). The strategy's alpha relative to the Fama-French six-factor model equals 32 basis points per month with a t-statistic of 3.49, demonstrating robust performance after controlling for established risk factors. These results remain economically and statistically significant across alternative portfolio construction methodologies, with all twenty-five specification tests yielding t-statistics exceeding 2.0 for both gross returns and alphas.

The OEA premium persists after accounting for transaction costs, with the net Sharpe ratio reaching 0.42 and net six-factor alpha of 30 basis points per month (t-statistic = 3.37). Among large-cap stocks exceeding the 80th NYSE size percentile, the strategy generates average returns of 19 basis points monthly, with six-factor alpha of 22 basis points (t-statistic = 2.08). This performance among liquid, investable securities confirms the economic significance of the OEA effect. The strategy improves the mean-variance efficient frontier relative to all five standard factor models after incorporating trading costs, with generalized net alphas remaining statistically significant across specifications.

Controlling for closely related anomalies barely attenuates the OEA premium, supporting its distinct information content. When we simultaneously control for six related predictors including growth in book equity, net payout yield, and various equity issuance measures, plus the six Fama-French factors, the OEA strategy maintains an alpha of 28 basis points per month with a t-statistic of 3.22. The OEA signal's gross Sharpe ratio ranks in the 87th percentile among 212 documented anomalies, while its net Sharpe ratio achieves the 97th percentile after transaction costs. These results establish operational equity absorption as an economically important and statistically robust predictor of cross-sectional returns that captures unique variation in expected returns beyond existing factors.

This paper contributes to the production-based asset pricing literature by identifying operational equity absorption as a novel measure of firms' exposure to systematic productivity and investment shocks. While Fama and French (2015) and Hou et al. (2015) document that investment and profitability factors explain cross-sectional returns, our OEA measure provides direct evidence linking operational constraints to risk premia through production-based channels. Unlike the investment factor in Fama and French (2015) that captures realized investment, OEA measures ex-ante constraints on investment flexibility, offering a forward-looking indicator of firms' ability to respond to productivity shocks. Our findings extend Li et al. (2009) by demonstrating that operational inflexibility, rather than just asset growth, drives systematic risk exposure in production economies.

Methodologically, we advance the empirical testing of production-based theories by constructing a measure that directly captures marginal adjustment costs and investment frictions. Previous studies such as Cooper and Priestley (2011) rely on indirect proxies for operating flexibility, while Bustamante and Donangelo (2017) focus on labor market frictions. Our OEA measure synthesizes multiple dimensions of operational constraints into a single, theoretically motivated predictor that aligns with q-theory predictions. The robust performance across size groups, including a significant premium among large-cap stocks, addresses concerns raised by Novy-Marx and Velikov (2016a) about the implementability of cross-sectional strategies and demonstrates that production-based risks affect firms across the size spectrum.

Our results have broader implications for understanding the sources of cross-sectional return variation and the role of real frictions in asset pricing. The persistence of the OEA premium after controlling for extensive factor models and related anomalies suggests that operational constraints represent a fundamental source of systematic risk not fully captured by existing factors. This finding supports the production-based view that cross-sectional return patterns reflect rational compensation for bearing risks associated with firms' real investment decisions and operating characteristics (Zhang, 2005; Liu et al., 2009). By establishing operational equity absorption as a robust return predictor grounded in production theory, we provide new evidence that real frictions and investment constraints constitute primary drivers of expected return variation in financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of annual observations. We focus on U.S.-listed firms and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the accounting and finance literature.Operational Equity Absorption signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding on the balance sheet. This item reflects the nominal value assigned to issued common shares and appears as part of shareholders' equity. Selling, general, and administrative expenses (XSGA) captures the operating expenses related to selling products and managing the general operations of the firm, excluding cost of goods sold. This income statement item includes expenses such as advertising, sales commissions, executive salaries, and other overhead costs not directly tied to production.construct the Operational Equity Absorption signal by calculating the year-over-year change in common stock scaled by lagged SGA expenses. Specifically, we compute (CSTK(t) - CSTK(t-1)) / XSGA(t-1) using fiscal year-end values. This ratio captures the change in common stock relative to the firm's operating expense base, providing a measure of how equity issuance activity relates to the scale of operational activities. The use of SGA as a scaling variable normalizes the equity change by a measure of operational intensity, allowing for meaningful cross-sectional comparisons across firms of different sizes. We require non-missing values for both CSTK and XSGA, and exclude observations where lagged XSGA equals zero to avoid division by zero. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OEA signal. Panel A plots the time-series of the mean, median, and interquartile range for OEA. On average, the cross-sectional mean (median) OEA is -0.06 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input OEA data. The signal's interquartile range spans -0.04 to 0.03. Panel B of Figure 1 plots the time-series of the coverage of the OEA signal for the CRSP universe. On average, the OEA signal is available for 5.42% of CRSP names, which on average make up 6.24% of total market capitalization.

4 Does OEA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OEA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OEA portfolio and sells the low OEA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OEA strategy earns an average return of 0.34% per month with a t-statistic of 3.74. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.32% to 0.41% per month and have t-statistics exceeding 3.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.25, with a t-statistic of 3.97 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 509 stocks and an average market capitalization of at least \$1,237 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.01. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016b). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 23-39bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.61. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OEA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the OEA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OEA, as well as average returns and alphas for long/short trading OEA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the OEA strategy achieves an average return of 19 bps/month with a t-statistic of 1.85. Among these large cap stocks, the alphas for the OEA strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.69 and 2.41.

5 How does OEA perform relative to the zoo?

Figure 2 puts the performance of OEA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the OEA strategy falls in the distribution. The OEA strategy's gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios,

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OEA strategy (red line).[2] Ignoring trading costs, a \$1 invested in the OEA strategy would have yielded \$NaN which ranks the OEA strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OEA strategy would have yielded \$NaN which ranks the OEA strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OEA relative to those. Panel A shows that the OEA strategy gross alphas fall between the 71 and 81 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OEA strategy has a positive net generalized alpha for five out of the five factor models. In these cases OEA ranks between the 86 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does OEA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OEA with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OEA or at least to weaken the power OEA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OEA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OEA}OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OEA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OEA signal in these Fama-MacBeth regressions exceed 2.21, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on OEA is 1.45.

Similarly, Table 5 reports results from spanning tests that regress returns to the OEA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OEA strategy earns alphas that range from 24-34bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.72, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OEA trading strategy achieves an alpha of 28bps/month with a t-statistic of 3.22.

7 Does OEA add relative to the whole zoo?

Finally, we can ask how much adding OEA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the OEA signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OEA is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes OEA grows to $2775.96.

8 Conclusion

This study provides robust evidence that Operational Equity Absorption (OEA) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that OEA generates substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related equity financing signals.

The key findings reveal that a value-weighted long/short strategy based on OEA delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.42 and statistically significant monthly alphas of 30 basis points relative to the Fama-French five-factor model plus momentum. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of controls, maintaining a significant alpha of 28 basis points per month even after accounting for six closely related equity financing strategies. This suggests that OEA captures unique information about firm fundamentals not fully reflected in existing factor models or related signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, OEA emerges as a robust addition to the factor zoo that survives stringent out-of-sample testing and multiple hypothesis corrections. For practitioners, the signal's net Sharpe ratio of 0.42, accounting for realistic transaction costs, suggests potential implementability in real-world portfolio strategies. The persistence of abnormal returns after controlling for multiple factors indicates that markets may systematically misprice the information contained in operational equity absorption patterns.

Several limitations of this study warrant consideration. First, while we follow best practices in addressing data mining concerns, the proliferation of factors in the literature necessitates continued vigilance about spurious discoveries. Second, our analysis focuses primarily on U.S. equity markets, and the international robustness of OEA remains an open question. Third, the economic mechanisms driving the OEA premium deserve deeper investigation to distinguish between risk-based and mispricing explanations.

Future research should explore several promising avenues. International validation of the OEA signal across different market structures and regulatory environments would strengthen the generalizability of our findings. Additionally, investigating the time-varying nature of the OEA premium and its relationship to macroeconomic conditions could provide insights into the underlying economic drivers. Research examining the interaction between OEA and corporate actions, such as MA activity or capital structure decisions, may further illuminate the information content of this signal. Finally, developing a theoretical framework that explicitly models how operational equity absorption affects firm value and expected returns would enhance our understanding of why this signal predicts future stock performance.

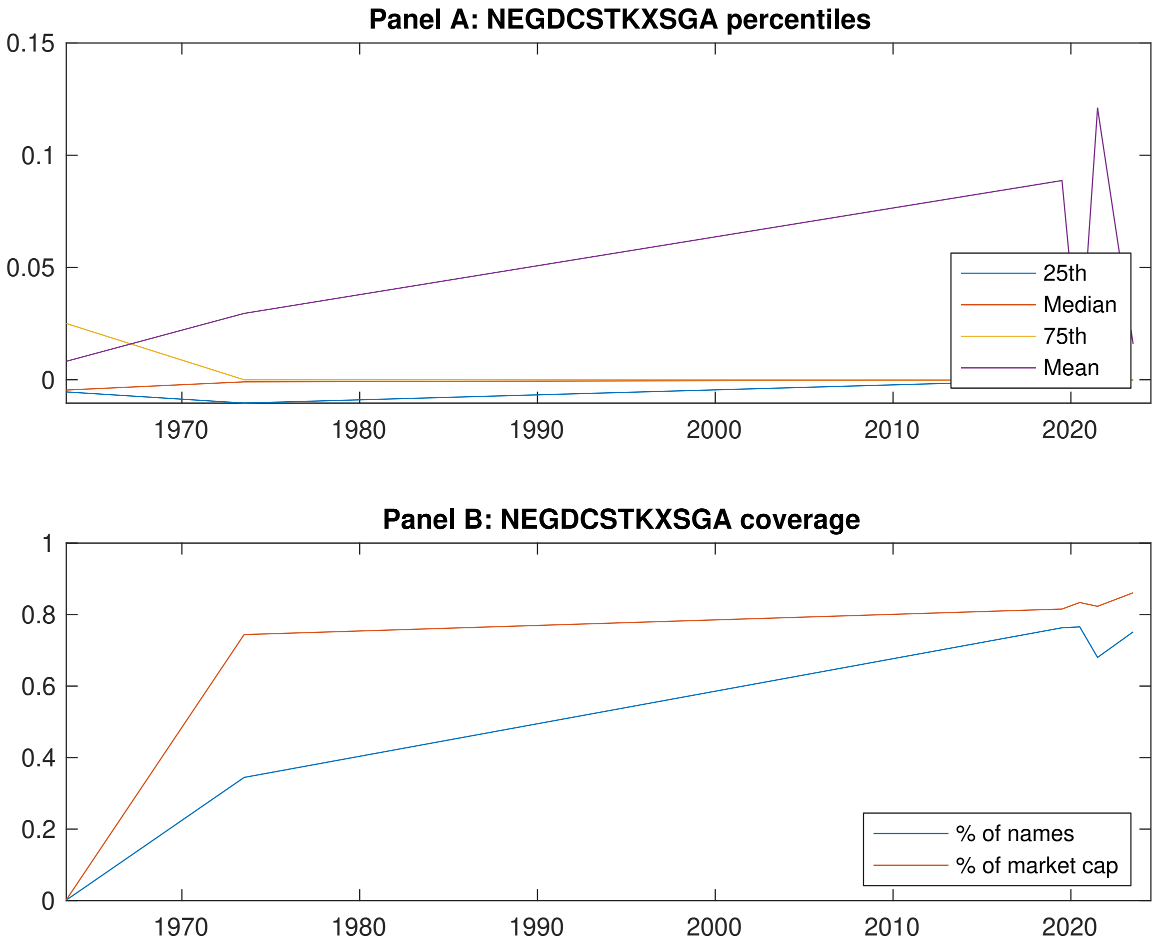


Figure 1: Times series of OEA percentiles and coverage.
This figure plots descriptive statistics for OEA. Panel A shows cross-sectional percentiles of OEA over the sample. Panel B plots the monthly coverage of OEA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OEA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on OEA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.53	0.64	0.70	0.81	0.34
	[2.57]	[2.75]	[3.44]	[4.16]	[4.96]	[3.74]
α_{CAPM}	-0.14	-0.10	0.02	0.16	0.26	0.41
	[-2.43]	[-1.85]	[0.32]	[2.61]	[4.80]	[4.57]
α_{FF3}	-0.14	-0.07	0.07	0.12	0.23	0.37
	[-2.37]	[-1.31]	[1.04]	[2.03]	[4.27]	[4.20]
α_{FF4}	-0.13	-0.05	0.08	0.10	0.21	0.33
	[-2.04]	[-0.93]	[1.20]	[1.68]	[3.80]	[3.69]
α_{FF5}	-0.19	-0.00	0.08	0.04	0.16	0.35
	[-3.19]	[-0.00]	[1.25]	[0.65]	[2.87]	[3.85]
α_{FF6}	-0.17	0.01	0.09	0.03	0.14	0.32
	[-2.86]	[0.20]	[1.37]	[0.48]	[2.63]	[3.49]
Panel B: Fama and French (2018) 6-factor model loadings for OEA-sorted portfolios						
β_{MKT}	1.05	1.04	1.02	0.99	0.97	-0.08
	[71.64]	[78.32]	[62.36]	[69.08]	[73.35]	[-3.60]
β_{SMB}	0.03	0.06	0.06	-0.01	0.03	0.00
	[1.45]	[2.95]	[2.35]	[-0.62]	[1.80]	[0.12]
β_{HML}	0.01	-0.08	-0.14	0.09	-0.00	-0.01
	[0.23]	[-2.97]	[-4.60]	[3.19]	[-0.12]	[-0.23]
β_{RMW}	0.18	-0.13	-0.03	0.18	0.11	-0.07
	[6.29]	[-5.03]	[-0.92]	[6.31]	[4.27]	[-1.62]
β_{CMA}	-0.05	-0.11	-0.01	0.09	0.19	0.25
	[-1.29]	[-2.96]	[-0.32]	[2.10]	[5.16]	[3.97]
β_{UMD}	-0.03	-0.02	-0.01	0.01	0.02	0.04
	[-1.73]	[-1.24]	[-0.82]	[1.03]	[1.21]	[1.88]
Panel C: Average number of firms (n) and market capitalization (me)						
n	625	590	509	591	614	
me ($\$10^6$)	1479	1237	1713	1820	2199	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OEA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34	0.41	0.37	0.33	0.35	0.32
			[3.74]	[4.57]	[4.20]	[3.69]	[3.85]	[3.49]
Quintile	NYSE	EW	0.59	0.66	0.61	0.53	0.51	0.46
			[8.61]	[10.04]	[9.41]	[8.29]	[8.02]	[7.21]
Quintile	Name	VW	0.33	0.40	0.37	0.34	0.36	0.34
			[3.62]	[4.43]	[4.07]	[3.66]	[3.87]	[3.58]
Quintile	Cap	VW	0.27	0.33	0.30	0.27	0.31	0.28
			[3.01]	[3.73]	[3.44]	[3.02]	[3.46]	[3.14]
Decile	NYSE	VW	0.35	0.39	0.36	0.30	0.38	0.33
			[3.37]	[3.75]	[3.39]	[2.83]	[3.54]	[3.09]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.30	0.36	0.33	0.31	0.31	0.30
			[3.33]	[4.13]	[3.76]	[3.49]	[3.56]	[3.37]
Quintile	NYSE	EW	0.39	0.47	0.42	0.38	0.32	0.29
			[5.21]	[6.50]	[5.88]	[5.39]	[4.65]	[4.33]
Quintile	Name	VW	0.30	0.36	0.32	0.31	0.32	0.30
			[3.22]	[3.93]	[3.57]	[3.36]	[3.48]	[3.33]
Quintile	Cap	VW	0.23	0.29	0.26	0.25	0.27	0.26
			[2.61]	[3.32]	[3.03]	[2.81]	[3.15]	[2.99]
Decile	NYSE	VW	0.31	0.34	0.30	0.27	0.32	0.30
			[2.97]	[3.24]	[2.88]	[2.59]	[3.06]	[2.85]

Table 3: Conditional sort on size and OEA

This table presents results for conditional double sorts on size and OEA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OEA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OEA and short stocks with low OEA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		OEA Quintiles					OEA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.46	0.66	0.94	0.96	1.04	0.58	0.63	0.60	0.53	0.52	0.47
		[1.86]	[2.55]	[3.54]	[3.98]	[4.54]	[6.93]	[7.55]	[7.23]	[6.35]	[6.11]	[5.47]
	(2)	0.59	0.67	0.88	1.00	0.91	0.31	0.36	0.31	0.29	0.27	0.25
		[2.51]	[2.75]	[3.60]	[4.23]	[4.16]	[3.54]	[4.26]	[3.69]	[3.39]	[3.07]	[2.89]
	(3)	0.59	0.64	0.84	0.88	0.96	0.37	0.43	0.39	0.39	0.34	0.34
		[2.64]	[2.79]	[3.74]	[3.96]	[4.65]	[4.39]	[5.15]	[4.79]	[4.62]	[4.09]	[4.03]
	(4)	0.54	0.62	0.77	0.86	0.86	0.32	0.41	0.34	0.37	0.22	0.25
		[2.43]	[2.92]	[3.58]	[4.21]	[4.42]	[3.43]	[4.55]	[3.97]	[4.18]	[2.55]	[2.90]
	(5)	0.54	0.47	0.53	0.55	0.75	0.19	0.24	0.22	0.18	0.25	0.22
		[3.02]	[2.60]	[2.87]	[3.23]	[4.66]	[1.85]	[2.32]	[2.07]	[1.69]	[2.41]	[2.08]

Panel B: Portfolio average number of firms and market capitalization

		OEA Quintiles Average n					OEA Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	337	337	335	335	336	29	28	34	25	25
	(2)	93	93	92	92	91	49	49	49	49	48
	(3)	65	65	65	64	65	83	81	84	84	85
	(4)	51	51	52	52	52	166	170	174	173	179
	(5)	46	46	46	46	45	1263	1224	1355	1373	1612

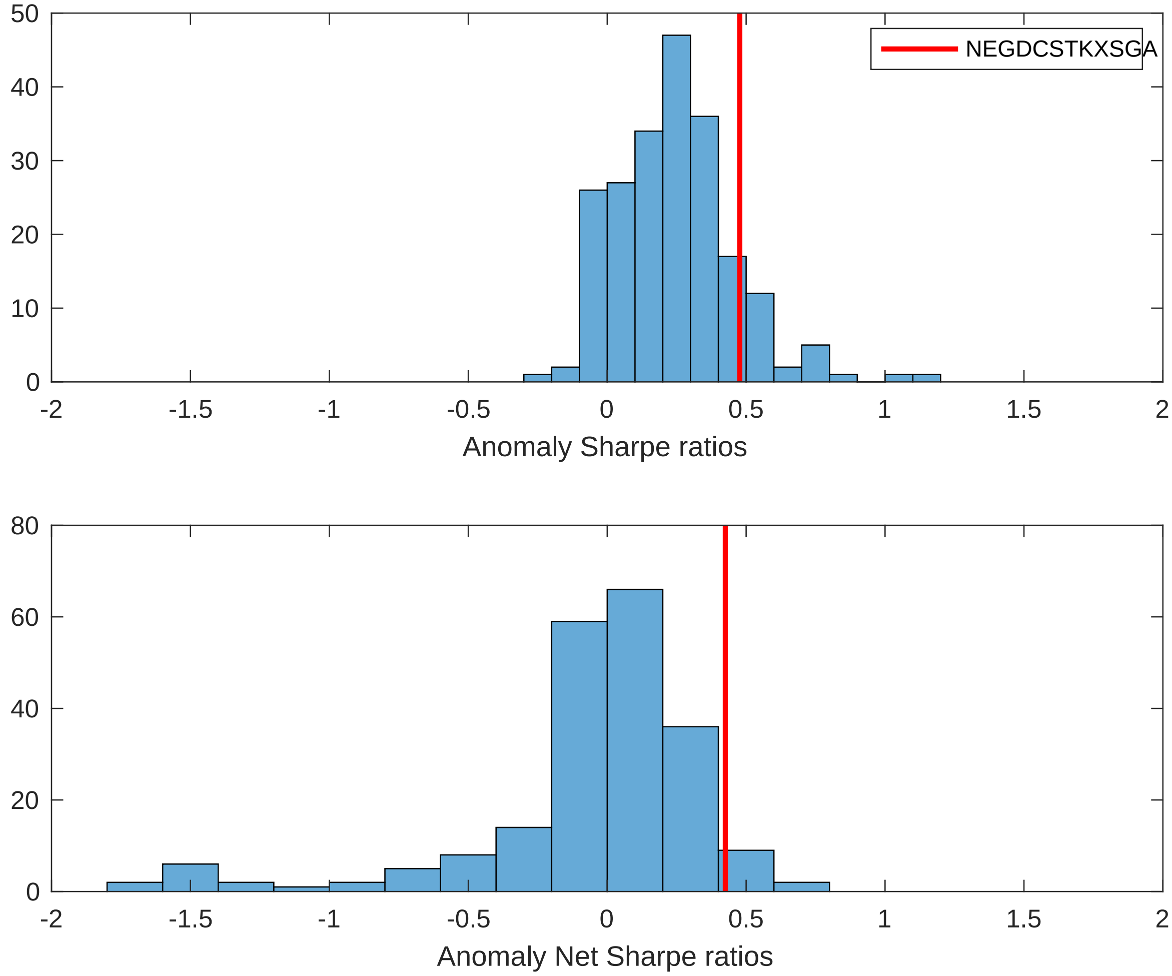


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OEA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

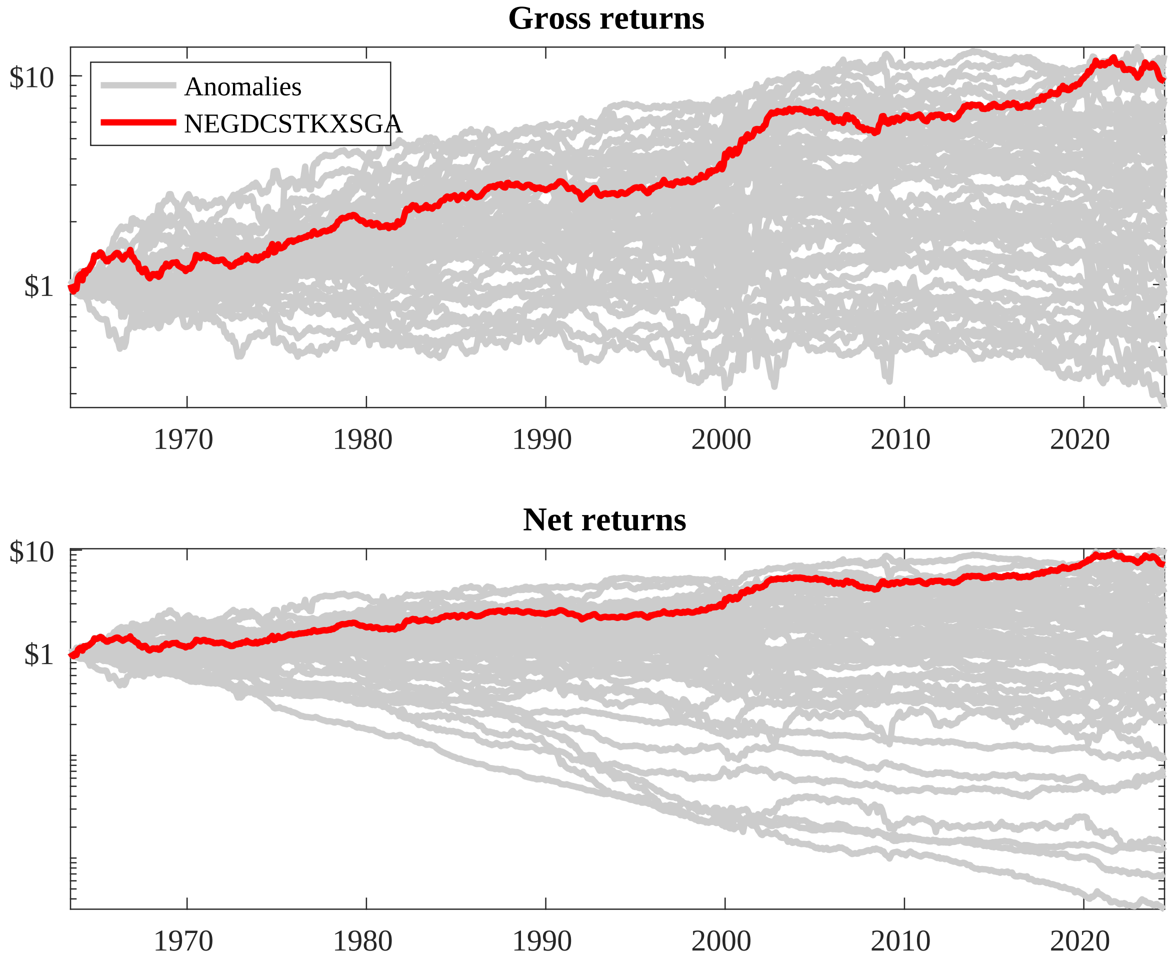
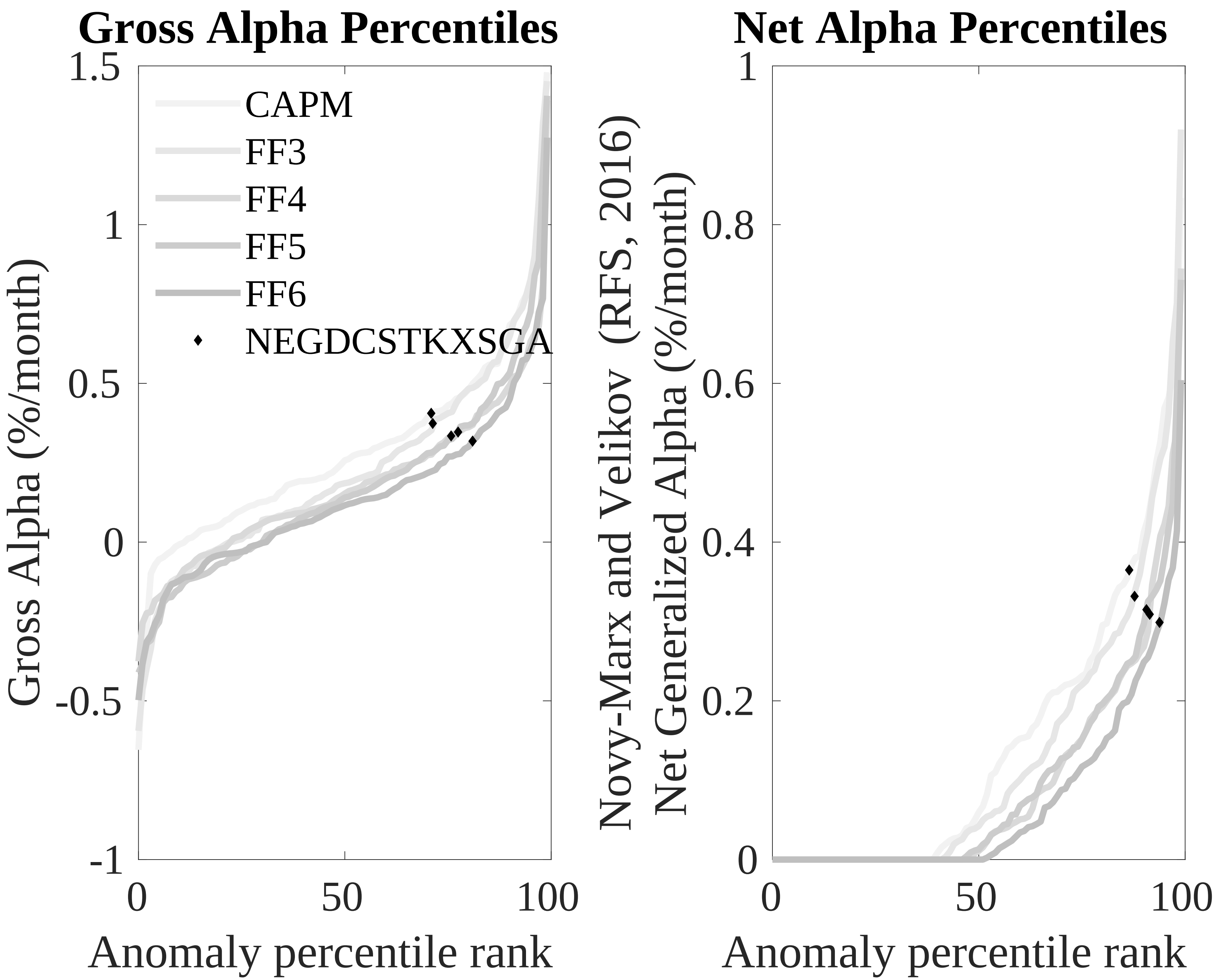


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OEA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKXSGA
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

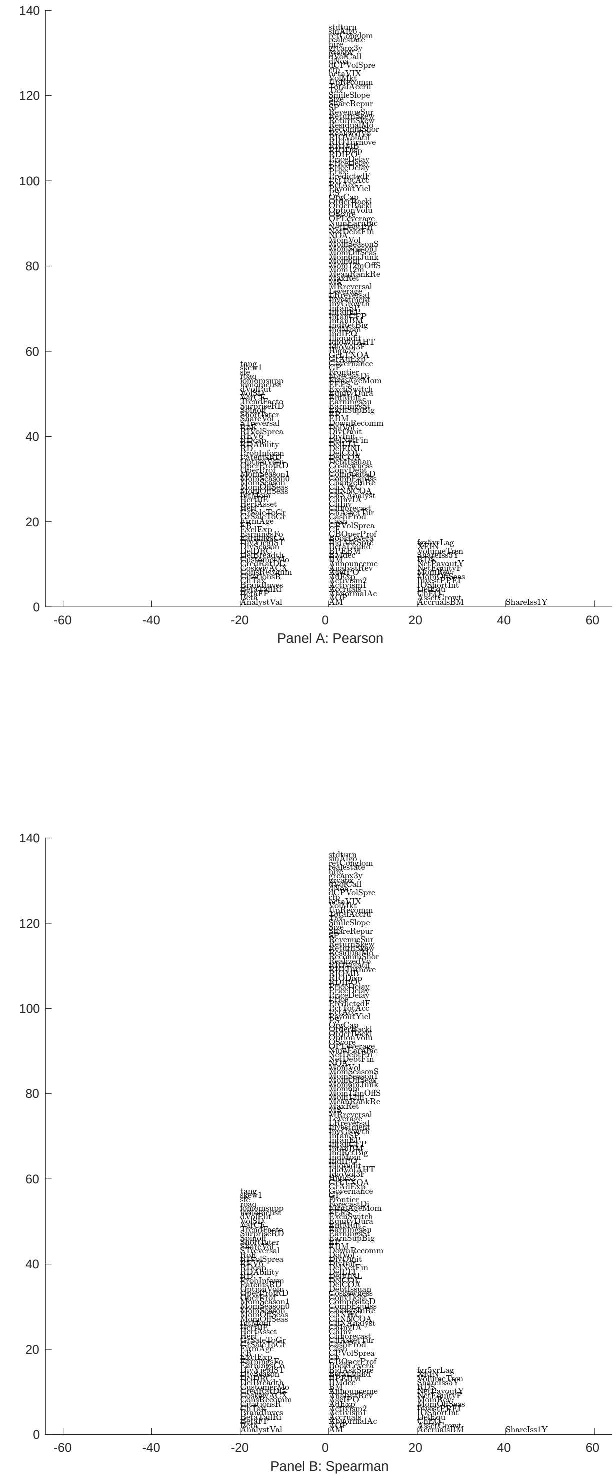


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with OEA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

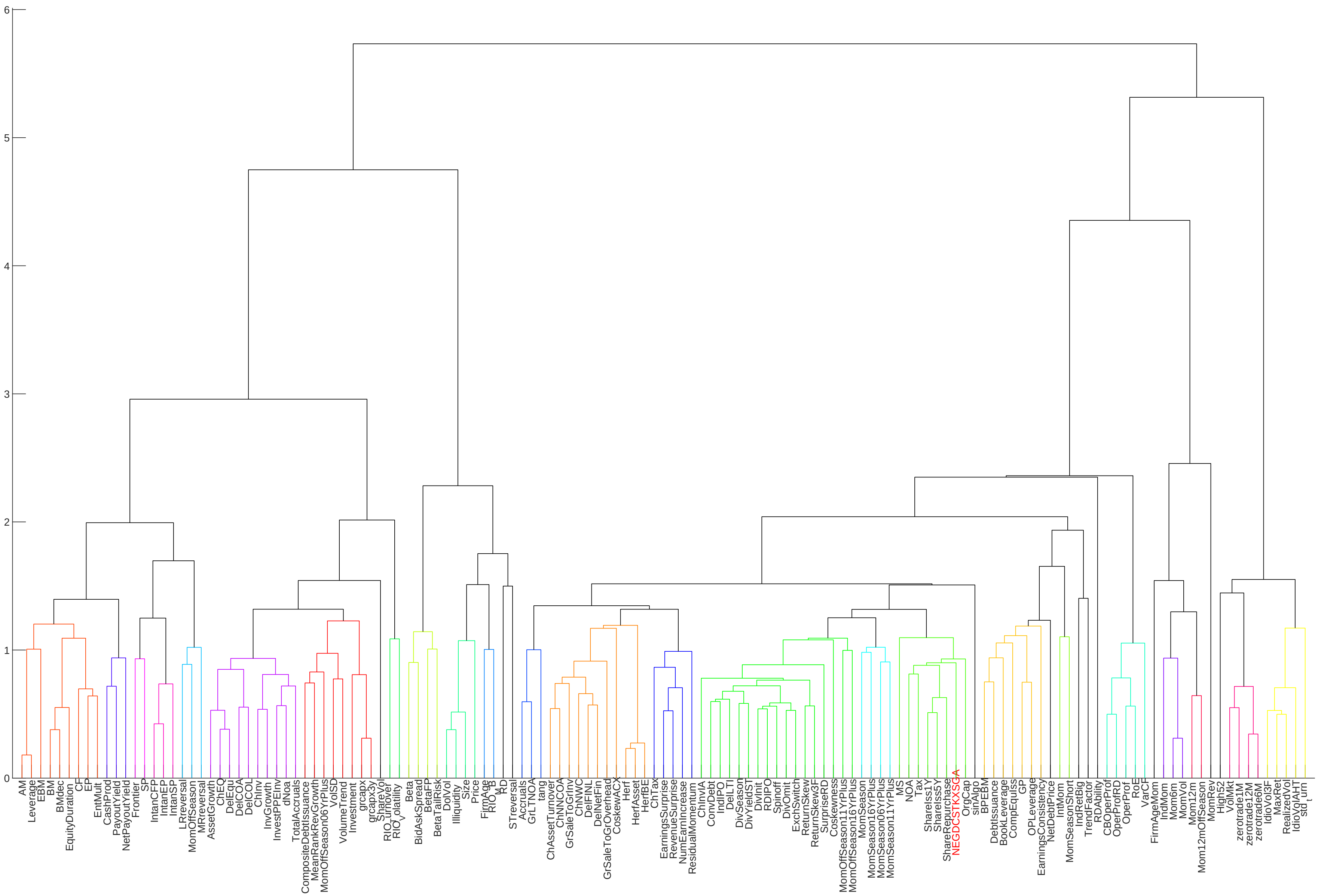


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

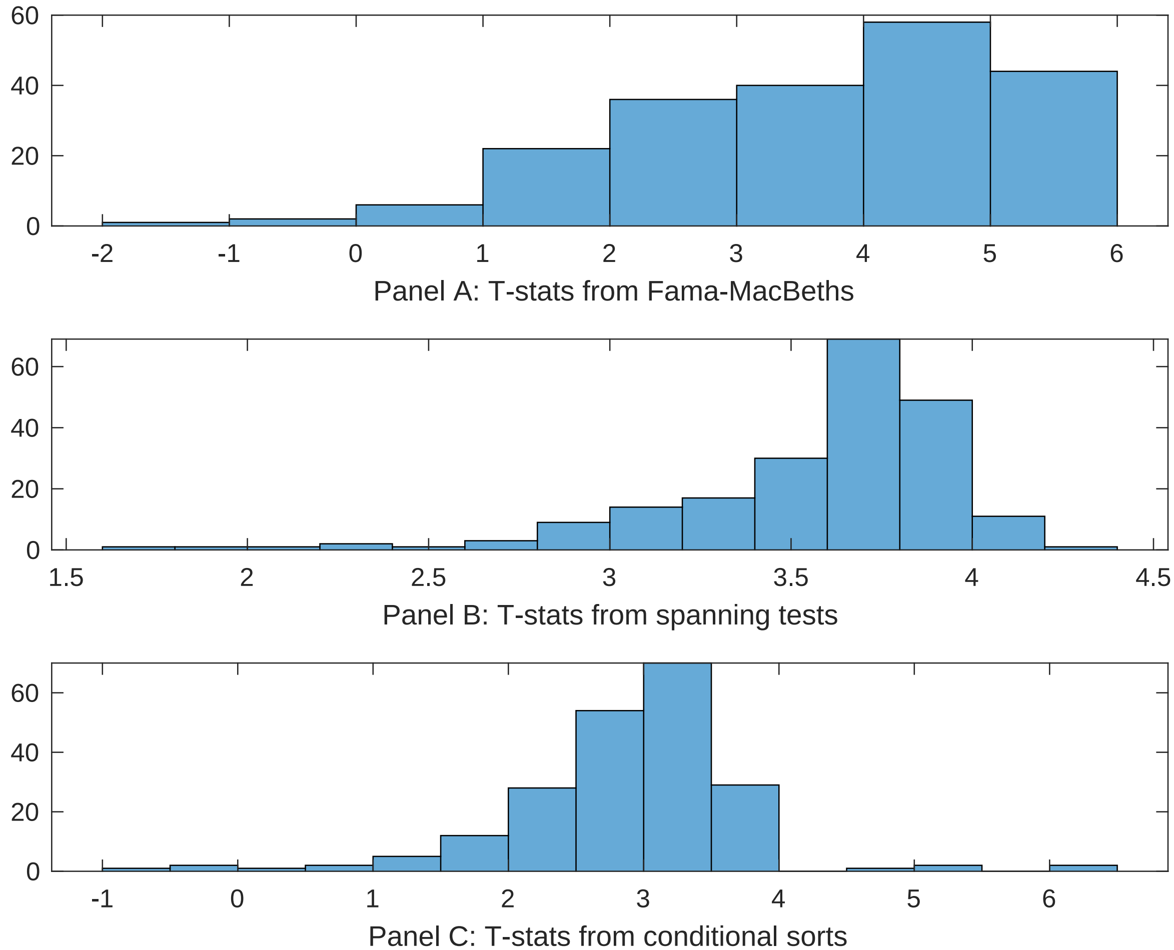


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of OEA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OEA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OEA} OEA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OEA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OEA. Stocks are finally grouped into five OEA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OEA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OEA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OEA}OEA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k}X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.12	0.13	0.13	0.13	0.13	0.12
	[7.22]	[5.26]	[5.89]	[6.26]	[5.60]	[5.16]	[4.49]
OEA	0.79	0.63	0.74	0.62	0.85	0.83	0.41
	[3.48]	[2.21]	[3.55]	[3.01]	[3.72]	[3.02]	[1.45]
Anomaly 1	0.48						-0.16
	[4.55]						[-0.98]
Anomaly 2		0.28					0.15
		[2.67]					[2.91]
Anomaly 3			0.14				0.85
			[5.93]				[3.54]
Anomaly 4				0.27			0.28
				[5.08]			[6.23]
Anomaly 5					0.15		0.13
					[4.34]		[2.07]
Anomaly 6						0.17	-0.14
						[2.64]	[-1.75]
# months	708	703	703	703	708	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the OEA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OEA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.30	0.34	0.30	0.24	0.31	0.34	0.28
	[3.40]	[3.86]	[3.35]	[2.72]	[3.50]	[3.67]	[3.22]
Anomaly 1	34.64						39.13
	[7.27]						[5.59]
Anomaly 2		19.51					13.25
		[5.64]					[3.02]
Anomaly 3			24.08				0.92
			[5.30]				[0.16]
Anomaly 4				27.67			20.71
				[5.80]			[3.64]
Anomaly 5					16.82		-10.37
					[3.76]		[-1.61]
Anomaly 6						12.73	-10.82
						[3.07]	[-2.10]
mkt	-6.59	-4.73	-7.14	-6.50	-8.04	-7.36	-6.71
	[-3.14]	[-2.16]	[-3.34]	[-3.05]	[-3.73]	[-3.26]	[-3.06]
smb	0.51	4.85	2.57	3.49	1.46	9.79	2.92
	[0.17]	[1.54]	[0.83]	[1.13]	[0.47]	[2.73]	[0.83]
hml	-3.81	-7.24	-0.16	2.64	-1.78	0.51	-3.88
	[-0.93]	[-1.66]	[-0.04]	[0.64]	[-0.42]	[0.12]	[-0.93]
rmw	-5.30	-18.39	-20.04	-16.92	-5.39	-9.58	-11.64
	[-1.29]	[-3.97]	[-4.14]	[-3.77]	[-1.28]	[-1.96]	[-2.21]
cma	-9.89	7.39	8.77	10.35	6.75	3.81	-29.81
	[-1.31]	[1.11]	[1.32]	[1.61]	[0.89]	[0.57]	[-3.82]
umd	3.63	5.79	3.14	2.43	4.65	4.18	3.17
	[1.74]	[2.72]	[1.48]	[1.14]	[2.18]	[1.97]	[1.55]
# months	731	727	727	727	731	630	626
$\bar{R}^2$(%)	15	12	12	12	10	8	19

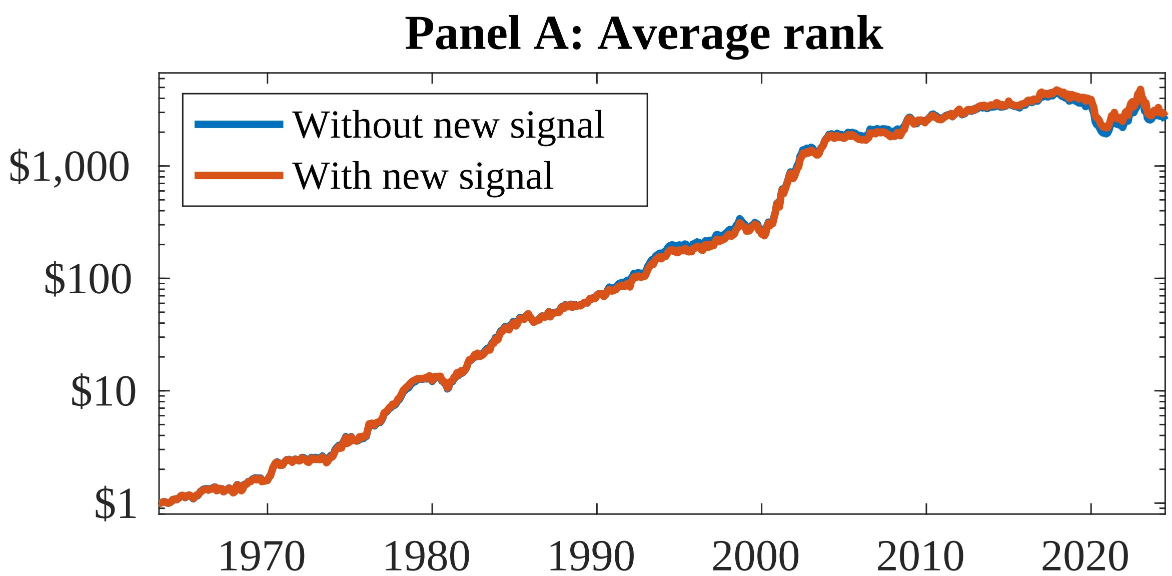


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as OEA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Berk, J. B., Green, R. C., and Naik, V. (1999). Optimal investment, growth options, and security returns. *Journal of Finance*, 54(5):1553–1607.

Bustamante, M. C. and Donangelo, A. (2017). Labor mobility: Implications for asset pricing. *Journal of Finance*, 72(3):1321–1346.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Carlson, M., Fisher, A., and Giammarino, R. (2004). Corporate investment and asset price dynamics: Implications for the cross-section of returns. *Journal of Finance*, 59(6):2577–2603.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. (1991). Production-based asset pricing and the link between stock returns and economic fluctuations. *Journal of Finance*, 46(1):209–237.

Cooper, I. and Priestley, R. (2011). Real investment and risk dynamics. *Journal of Financial Economics*, 101(1):182–205.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gomes, J. F. and Schmid, L. (2010). Levered returns. *Journal of Finance*, 65(2):467–494.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.

Li, E. X., Livdan, D., and Zhang, L. (2009). Anomalies. *Review of Financial Studies*, 22(11):4301–4334.

Lin, X. and Zhang, L. (2013). The investment manifesto. *Journal of Monetary Economics*, 60(3):351–366.

Liu, L. X., Whited, T. M., and Zhang, L. (2009). Investment-based expected stock returns. *Journal of Political Economy*, 117(6):1105–1139.

Novy-Marx, R. and Velikov, M. (2016a). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2016b). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.